T. C.
ISTANBUL BEYKENT UNİVERSİTY
FACULTY OF ENGİNEERİNG AND ARCHİTECTURE
DEPARTMENT OF COMPUTER ENGİNEERİNG

CAMPUS ASSİSTANT

Mobile Application Programming

Project Proposal Report

Prepared by- Student Name

TÜRKÜ ÇAĞLAR - 2103021261

RUKİYE EBRAR GERÇEK - 2103021220

MELİS ÖZYAZICI - 2103021257

DENİZ UMUT BAYRAK -2103021223

KAAN YİĞİT ARSLAN – 2103021218

Supervisor Name

Onur KARDEŞ

December 2025

1. Introduction

Campus life requires students to manage various academic and daily responsibilities. However, important information—such as schedules, meals, events, and grade expectations—often exists across scattered and disconnected platforms.

This proposal introduces **Campus Assistant**, a mobile application developed with **Flutter**, aiming to unify all essential student tools within a single, intuitive interface.

2. Purpose of the Application

The main purpose of **Campus Assistant** is to simplify daily academic tasks for university students by offering:

- Fast access to **weekly class schedules**
- **Daily and weekly cafeteria menus**
- A clear overview of **upcoming campus events**
- An intuitive **grade calculator** that outputs both numeric and letter grades
- A clean and centralized campus productivity experience

This prototype is designed specifically for **Beykent University** students but has the potential to expand into a general-purpose student companion app.

3. Target Users

3.1. Primary Users

- Undergraduate and graduate students who need quick access to university-related information.

3.2. Secondary Users

- Freshmen adjusting to the academic environment
- Student club members who follow campus events

The app is designed to enhance convenience and productivity for all students.

4. Overall Design & Major Functionalities

Campus Assistant consists of **five core screens**, accessible via a bottom navigation bar.

The UI prioritizes clarity, simplicity, and professional design principles.

4.1. Home Screen

- Displays **current date** using Flutter's `intl` package
- Includes a friendly welcome message
- Offers quick-access cards to the main features
- Shows a lightweight **daily summary** of workload
- Provides a visually structured entry point to the app

Design Goal: Create a welcoming and fast-access dashboard that sets the tone for the entire user experience.

4.2. Schedule Screen

- Shows real **weekly class schedule** based on OBS data provided
- Organized into weekday tabs
- **Automatically selects today's tab** using system date
- Includes course name, times, and instructor info
- Future-ready structure for OBS API integration

Design Goal: Help students instantly see today's and upcoming classes without searching through multiple pages.

4.3. Cafeteria Screen

- Two views: **Today** and **This Week**
- Includes an interactive user field:
  "Today I plan to eat…"
- Weekly menu displayed through expandable cards
- Uses Material components to maintain simplicity and readability

Design Goal: Provide both structured cafeteria information and a personalized user interaction point.

4.4. Events Screen

- Lists upcoming campus events (e.g., seminars, club events, workshops).
- Each event includes a **category tag** (Career, Technology, Social).
- Uses card layouts for clean readability.

Design Goal: Enhance campus engagement by giving students an easy way to explore activities.

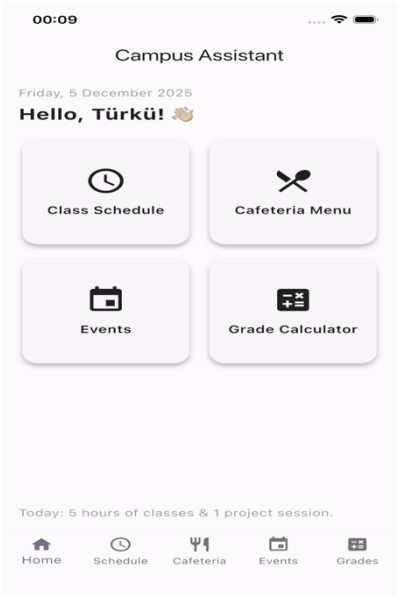
4.5. Grade Calculator Screen

- Inputs: Midterm grade + Final grade
- Outputs:
  - Weighted final score
  - Letter grade (AA–FF)
- Uses standard Turkish university grading scale
- Interactive and clean layout for fast use

Design Goal: Provide a reliable and fast way for students to estimate academic performance.

5. Mockup Screen Designs

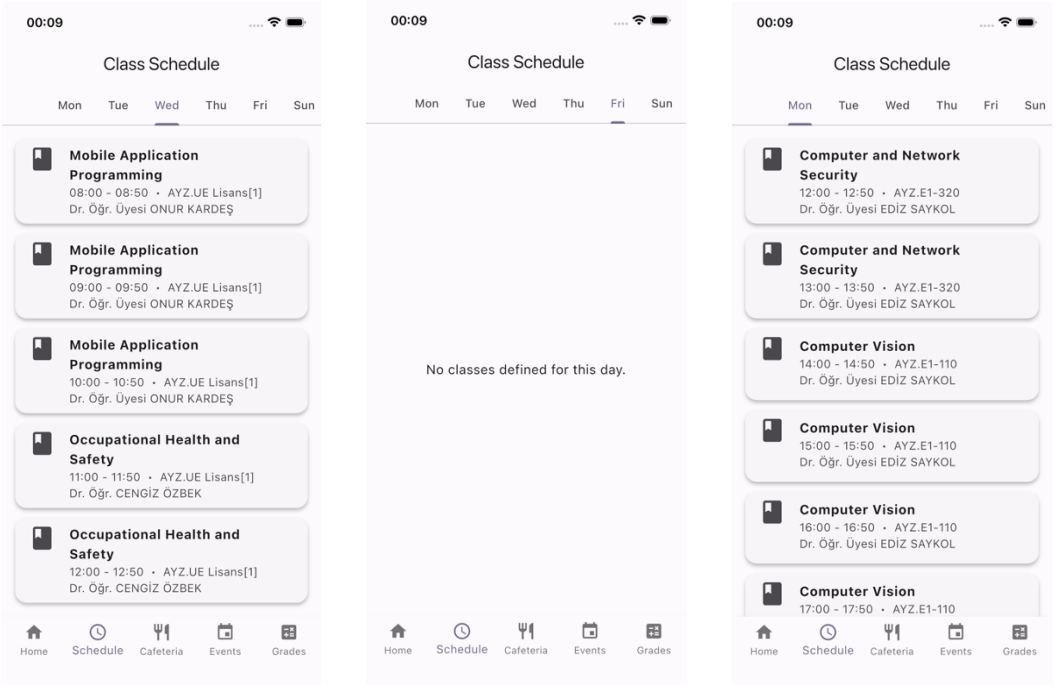
5.1. Home Screen Mockup



Description:

- Displays current date and personalized greeting
- Four quick-access navigation cards
- Clean white background for academic look

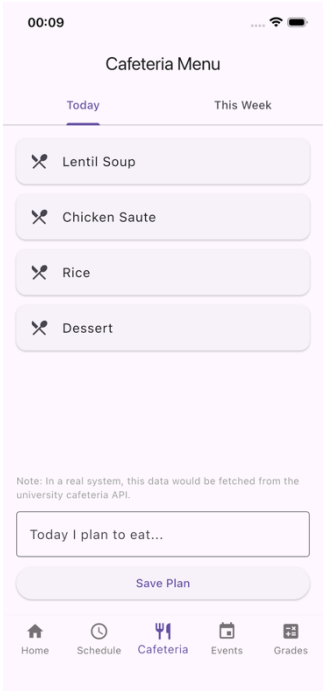
5.2. Schedule Screen Mockup

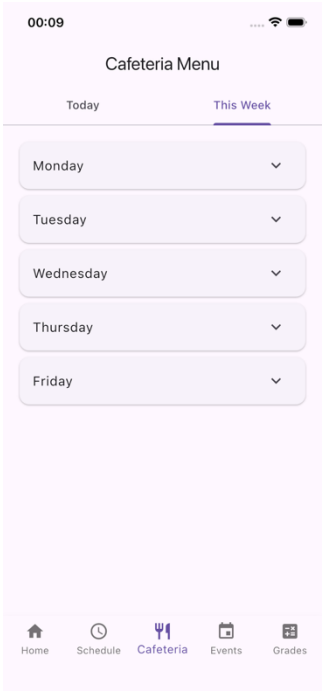


Description:

- Weekly tab layout
- Today's schedule highlighted through auto-selection
- Course cards include time + instructor data

5.3. Cafeteria Menu Screen Mockup

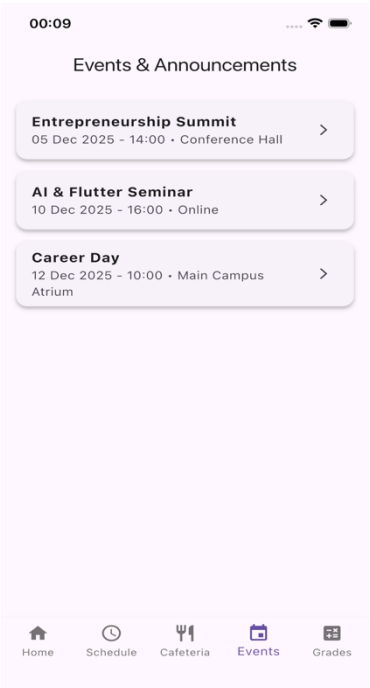




Description:

- Today's meals listed as cards
- User input section ("Today I plan to eat…")
- Weekly menu displayed through structured expansion tiles

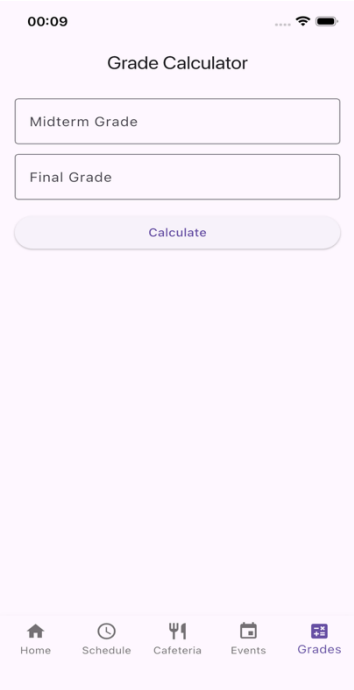
5.4. Events Screen Mockup



Description:

- Event titles + dates + short descriptions
- Category chips for filtering future versions
- Card-based layout for readability

5.5. Grade Calculator Mockup



Description:

- Clean input fields
- Numeric and letter grade output
- Simple interaction flow

6. Future Improvements

- Push notifications for upcoming classes
- Assignment/task manager
- Dark mode support
- Integration with OBS student information system
- Real-time cafeteria API
- Campus map with building navigation
- User accounts & profile customization

7. Conclusion

Campus Assistant effectively consolidates all essential campus tools into one modern and simple mobile interface. The application prototype already demonstrates major functionality through real Flutter development and represents a scalable foundation for future enhancements.

This project not only meets the requirements of the Mobile Application Programming course but also reflects a real-world student companion application that could be deployed across campus.